

LIMITS OF HIGHER COHOMOLOGY OF LINE BUNDLES

ABSTRACT. Let X be a smooth complex projective variety of pure dimension n and let L be a line bundle on X . We prove that $k^{-n}h^j(X, L^{\otimes k})$ has a finite limit for every cohomological degree j . A bound in terms of regularity, support dimension, and sections at a fixed twist controls Frobenius direct images on an arithmetic model. This gives a comparison between exponents k and pk with an error uniform in the prime p . Upper semicontinuity and the prime number theorem yield convergence in characteristic zero.

1. INTRODUCTION

Let X be a smooth complex projective variety of dimension n , and let L be a line bundle on X . Write $h^j(X, \mathcal{F}) = \dim_{\mathbb{C}} H^j(X, \mathcal{F})$. The asymptotic cohomological functions are defined by

$$\widehat{h}^j(X, L) = \limsup_{k \rightarrow \infty} \frac{h^j(X, L^{\otimes k})}{k^n/n!}.$$

These functions were introduced and studied by Küronya [Kö6, p. 1476]. The existence of ordinary limits in positive degree is identified as an open problem in [Li26, §1]. We prove convergence for smooth complex projective varieties.

Theorem A. *Let X be a smooth complex projective variety of pure dimension n , and let L be a line bundle on X . For every integer $j \geq 0$, the finite limit*

$$\lim_{k \rightarrow \infty} \frac{h^j(X, L^{\otimes k})}{k^n}$$

exists.

Thus the limsup defining $\widehat{h}^j(X, L)$ can be replaced by a limit. We use the normalization k^n throughout the proof. A holomorphic line bundle on a projective manifold is algebraic, and its algebraic and analytic cohomology agree by [Ser56, §3, no. 12, Theorems 1–3]; hence the theorem also applies in the analytic formulation.

The comparison needed for the proof concerns varying sheaves on varying reductions of X . A bound for each fixed sheaf would allow its constant to depend on the characteristic and would not suffice. **Theorem 2.2** gives a bound linear in the dimension of sections at a fixed twist, with all other dependencies explicit. For a Frobenius direct image of rank p^n , a fixed twist generates the sheaf and the resulting quotient has support of dimension at most $n - 1$. This yields the uniform estimate in **Theorem 3.2**. After specialization, it gives an upper bound for normalized cohomology at every sufficiently large prime multiple of any fixed exponent. The bound for consecutive exponents in **Theorem 4.1** and the prime number theorem extend that upper bound to all large exponents.

The use of reductions in many prime characteristics followed by the prime number theorem also occurs in Totaro’s proof of the equivalence of cohomological definitions of partial ampleness [Tot13, Theorem 6.3 and its proof, pp. 744–746]. Here the positive-characteristic input is a quantitative comparison of cohomology dimensions. All sheaves below are coherent. Over any ground field, h^j denotes cohomology dimension over that field. Tensor products are over the structure sheaf, and L^k abbreviates $L^{\otimes k}$. We set $h^{-1} = 0$ when using long exact cohomology sequences.

2. COHOMOLOGY BOUNDS AT THE SUPPORT DIMENSION

Let Y be a projective scheme of dimension $n \geq 1$ over an algebraically closed field, let A be a very ample line bundle, and let L be a line bundle. A coherent sheaf $\mathcal{F}$ is m -regular with respect

to A if

$$H^i(Y, \mathcal{F} \otimes A^{m-i}) = 0 \quad (i > 0).$$

Applying Castelnuovo–Mumford regularity to the pushforward under the embedding defined by A , m -regularity implies $(m+1)$ -regularity and global generation of $\mathcal{F} \otimes A^m$ [Stacks, Tag 08A2, Lemmas 33.35.10–33.35.12]. In particular, higher cohomology vanishes after every twist at least m . We use an integer regularity bound rather than the least such integer, which need not exist for a sheaf with zero-dimensional support.

Choose an exact sequence

$$(2.1) \quad 0 \longrightarrow \mathcal{K}_n \longrightarrow \mathcal{E}_{n-1} \longrightarrow \cdots \longrightarrow \mathcal{E}_0 \longrightarrow L \longrightarrow 0, \quad \mathcal{E}_i = (A^{-b_i})^{\oplus r_i},$$

whose successive kernels are locally free. To construct it, take a sufficiently positive twist that generates L , then repeat this construction for each kernel, using ample global generation [Stacks, Tag 01PR, Proposition 28.27.13]. Each surjection onto a locally free sheaf splits locally, so its kernel is locally free. Fix nonnegative integers c and b such that

$$(2.2) \quad c \geq \max\{0, b_i - i : 0 \leq i < n\}, \quad B = A^b \otimes L^{-1} \text{ is globally generated.}$$

Lemma 2.1. *If $\mathcal{F}$ is m -regular, then $\mathcal{F} \otimes L^k$ is $(m + ck)$ -regular for every integer $k \geq 0$.*

Proof. Tensoring (2.1) with $\mathcal{F}$ preserves exactness because the short exact sequences defining its kernels split locally. For $j > 0$, twist this sequence by A^{m+c-j} . The cohomology groups of its free terms that occur in dimension shifting are

$$H^{j+i}(Y, \mathcal{F} \otimes A^{m+c-j-b_i}), \quad 0 \leq i < n.$$

They vanish: if $j+i \leq n$, the twist satisfies $m+c-j-b_i \geq m-(j+i)$, and monotonicity of regularity applies; if $j+i > n$, cohomology vanishes by dimension. The terminal group $H^{j+n}(Y, \mathcal{F} \otimes \mathcal{K}_n \otimes A^{m+c-j})$ vanishes for the same dimension reason [Stacks, Tag 0A4R]. The long exact sequences therefore give $H^j(Y, \mathcal{F} \otimes L \otimes A^{m+c-j}) = 0$. Iterating this assertion proves the lemma. $\square$

Proposition 2.2. *Fix integers $0 \leq d \leq n$ and $T \geq s \geq 0$. There is a constant C , depending only on n, d, s, T, c, b , such that, for every real $M \geq 0$, each s -regular coherent sheaf $\mathcal{F}$ satisfying*

$$\dim \operatorname{Supp} \mathcal{F} \leq d, \quad h^0(Y, \mathcal{F} \otimes A^T) \leq M$$

obeys

$$(2.3) \quad h^j(Y, \mathcal{F} \otimes L^k \otimes A^t) \leq CM(k+t+1)^d \quad (j, k, t \geq 0).$$

In particular, the same constant applies on different projective schemes over different algebraically closed fields when the displayed integer data, the resolution (2.1), and the global generation in (2.2) are available with the same bounds.

Proof. The zero sheaf satisfies the assertion. We argue by induction on d , simultaneously for all j, k, t . If $d = 0$, higher cohomology vanishes and the dimension of global sections is the length of $\mathcal{F}$, unchanged by tensoring with any line bundle. It is at most M .

Suppose $d > 0$. Each associated point of $\mathcal{F}$ excludes a proper linear subspace of $H^0(Y, A)$, since A is globally generated. The field is infinite, so choose a section outside their finite union. Multiplication by this section is injective. Its zero scheme meets every positive-dimensional component of $\operatorname{Supp} \mathcal{F}$ properly and avoids the zero-dimensional components. Thus it gives

$$(2.4) \quad 0 \longrightarrow \mathcal{F} \otimes A^{-1} \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow 0, \quad \dim \operatorname{Supp} \mathcal{G} \leq d-1.$$

The cohomology sequence at twist $s-i$ shows that $\mathcal{G}$ is s -regular. Moreover, $H^1(Y, \mathcal{F} \otimes A^{T-1}) = 0$ because $T \geq s$, so

$$h^0(Y, \mathcal{G} \otimes A^T) \leq h^0(Y, \mathcal{F} \otimes A^T) \leq M.$$

The induction hypothesis applies to $\mathcal{G}$ with the same s, T, M .

We first estimate $h^0(Y, \mathcal{F} \otimes A^v)$ for $v \geq 0$. If $v \leq T$, a power of the chosen section injects $\mathcal{F} \otimes A^v$ into $\mathcal{F} \otimes A^T$, giving the bound M . For $v > T$, (2.4) gives

$$h^0(Y, \mathcal{F} \otimes A^v) \leq h^0(Y, \mathcal{F} \otimes A^{v-1}) + h^0(Y, \mathcal{G} \otimes A^v).$$

The last term is at most $C_{d-1}M(v+1)^{d-1}$ by induction. Summation yields a constant C'_d with

$$(2.5) \quad h^0(Y, \mathcal{F} \otimes A^v) \leq C'_d M(v+1)^d \quad (v \geq 0).$$

Choose also a section of the globally generated line bundle B avoiding the associated points of $\mathcal{F}$. Its k th power gives an injection

$$\mathcal{F} \otimes L^k \otimes A^t \longrightarrow \mathcal{F} \otimes A^{bk+t}.$$

Estimate (2.5) proves (2.3) when $j = 0$.

For $j > 0$, set $U = \max\{0, s + ck - t\}$. By Theorem 2.1,

$$H^j(Y, \mathcal{F} \otimes L^k \otimes A^{t+U}) = 0.$$

Successive twists of (2.4) then give

$$h^j(Y, \mathcal{F} \otimes L^k \otimes A^t) \leq \sum_{\ell=1}^U h^{j-1}(Y, \mathcal{G} \otimes L^k \otimes A^{t+\ell}).$$

There are at most $s + ck$ summands, and $k + t + \ell + 1 \leq C_0(k + t + 1)$ with C_0 depending only on s, c . Applying the induction hypothesis gives (2.3). The constants in these recurrences use only the integers in the statement. The chosen sections may vary with the sheaf and the field, but their choices do not enter the constants. $\square$

3. UNIFORM COMPARISON UNDER FROBENIUS

We first fix an arithmetic model, including the data needed in Section 2. All subsequent uniformity assertions refer to this one model.

Lemma 3.1. *Let X be a connected smooth complex projective variety of dimension $n \geq 1$, let A be a very ample line bundle, and let L be a line bundle. There are an integral finitely generated $\mathbb{Z}$ -subalgebra $R \subset \mathbb{C}$, a smooth projective morphism $\pi : \mathcal{X} \rightarrow \operatorname{Spec} R$ of relative dimension n with geometrically integral fibers, and line bundles $\mathcal{A}, \mathcal{L}$ recovering A, L after base change to $\mathbb{C}$, with the following properties. The line bundle $\mathcal{A}$ is relatively very ample; the resolution (2.1) is available on every fiber with the same shifts and locally free kernels; $\mathcal{A}^b \otimes \mathcal{L}^{-1}$ is relatively globally generated for one fixed $b \geq 0$; and there are integers $s_0 \geq 0$ and $M_0 \geq 1$ such that every geometric fiber Y satisfies*

$$\mathcal{O}_Y \text{ is } s_0\text{-regular,} \quad h^0(Y, \mathcal{A}_Y^{s_0}) \leq M_0.$$

For any fixed nonnegative integers q, t , the dimensions $h^q(Y, \mathcal{A}_Y^t)$ are bounded on this model.

Proof. Choose a projective embedding and the finite resolution on X constructed in (2.1). The equations, transition functions, morphisms, and finite sets of generators for these data involve finitely many complex coefficients. They descend to an integral finitely generated $\mathbb{Z}$ -subalgebra of $\mathbb{C}$; finite presentation descent for schemes and sheaves is recorded in [Stacks, Tags 01ZM and 01ZR]. After localization, the model is smooth of relative dimension n and the embedding is relatively very ample. Its generic fiber is geometrically integral because its base change to $\mathbb{C}$ is X . Geometric connectedness persists on a nonempty open [Stacks, Tag 055C, Lemma 37.28.4]; smooth geometrically connected fibers are geometrically integral, since irreducible components of a regular scheme are disjoint.

Descend the short exact sequences with their locally free kernels rather than only the maps in the resolution. Local freeness, exactness, and surjectivity of the chosen evaluation maps hold on the generic fiber. Their failure loci are closed in the projective model, so their images in the base are closed and miss the generic point. Removing those images makes the descended sequences exact with locally free kernels and makes the evaluation maps surjective. The short exact sequences split locally and therefore remain exact on every fiber. This also retains global generation of $\mathcal{A}^b \otimes \mathcal{L}^{-1}$ with one fixed b .

Serre vanishing on the generic fiber supplies an integer $s_0 \geq 0$ for which the finitely many groups $H^i(\mathcal{O}(s_0 - i))$, $1 \leq i \leq n$, vanish [Stacks, Tag 01XO]. Upper semicontinuity preserves these vanishings after a further localization. The same semicontinuity theorem, applied to each

fixed twist, bounds the dimensions of sections and the other fixed cohomology groups on the quasi-compact base [Stacks, Tag 0BDN]. In particular, it gives M_0 . These choices are made once; none depends on a subsequently chosen exponent or prime. $\square$

For an exact sequence $0 \rightarrow \mathcal{U} \rightarrow \mathcal{V} \rightarrow \mathcal{W} \rightarrow 0$, integer regularity bounds $r_{\mathcal{U}}, r_{\mathcal{V}}$ give the bound

$$(3.1) \quad r_{\mathcal{W}} = \max\{r_{\mathcal{V}}, r_{\mathcal{U}} - 1\}.$$

Indeed, the groups adjacent to $H^i(\mathcal{W} \otimes A^{r_{\mathcal{W}}-i})$ are $H^i(\mathcal{V} \otimes A^{r_{\mathcal{W}}-i})$ and $H^{i+1}(\mathcal{U} \otimes A^{r_{\mathcal{W}}-i})$, both zero by monotonicity of regularity.

Proposition 3.2. *On a fixed model as in Theorem 3.1, there is a constant C such that, for every geometric fiber Y in characteristic $p > 0$, every integer $k \geq 0$, and every $j \geq 0$,*

$$(3.2) \quad |h^j(Y, L_Y^{p^k}) - p^n h^j(Y, L_Y^k)| \leq Cp^n(k+1)^{n-1},$$

where $L_Y = \mathcal{L}|_Y$. The constant is independent of p , the fiber, k , and j .

Proof. Write $A_Y = \mathcal{A}|_Y$ and suppress subscripts. Take the absolute Frobenius $F : Y \rightarrow Y$ on this geometric fiber. Over its algebraically closed, hence perfect, ground field, F is finite flat of degree p^n . Finiteness and degree follow from [Stacks, Tag 0CC6, Lemmas 33.36.8 and 33.36.10], and flatness follows from [Stacks, Tag 0FK6, Lemma 50.19.5]. The relative and absolute Frobenius differ by the isomorphism induced by Frobenius on the perfect ground field. Thus $\mathcal{E} = F_*\mathcal{O}_Y$ is locally free of rank $r = p^n$.

Set

$$u = s_0 + n + 1, \quad S = s_0 + u, \quad T = S.$$

These integers are fixed on the model. The projection formula [Stacks, Tag 01E6, Lemma 20.54.2] and finiteness of F give, in dimension,

$$h^i(Y, \mathcal{E} \otimes A^{u-i}) = h^i(Y, A^{p(u-i)}) = 0 \quad (1 \leq i \leq n),$$

because $p(u-i) \geq s_0$. Degrees above n vanish by dimension. Hence $\mathcal{E}$ is u -regular, and $\mathcal{E} \otimes A^u$ is globally generated. Choose r of its global sections forming a basis at the generic point. They define an exact sequence

$$(3.3) \quad 0 \rightarrow (A^{-u})^{\oplus r} \rightarrow \mathcal{E} \rightarrow \mathcal{Q} \rightarrow 0, \quad \dim \text{Supp } \mathcal{Q} \leq n-1.$$

The first map is generically an isomorphism; its kernel is zero because its source is torsion-free on the integral scheme Y . The support assertion follows from generic surjectivity.

The source in (3.3) is $(s_0 + u)$ -regular. Formula (3.1) shows that $\mathcal{Q}$ is S -regular. Since $T - u = s_0$, the higher cohomology of the source at twist T vanishes, and in particular

$$h^0(Y, \mathcal{Q} \otimes A^T) \leq h^0(Y, \mathcal{E} \otimes A^T) = h^0(Y, A^{pT}).$$

The pure ample bound (2.5), applied to $\mathcal{O}_Y$ with regularity and reference twist both equal to s_0 and reference bound M_0 , gives $h^0(Y, A^{pT}) \leq C_0 p^n$ for the fixed integer $T = S$. Here C_0 is fixed on the model. Applying Theorem 2.2 to $\mathcal{Q}$ with $d = n-1$ and $M = C_0 r$ yields

$$h^i(Y, \mathcal{Q} \otimes L^k) \leq C_1 r(k+1)^{n-1} \quad (i, k \geq 0).$$

The long exact sequence of (3.3), after tensoring with L^k , now gives

$$(3.4) \quad |h^j(Y, \mathcal{E} \otimes L^k) - r h^j(Y, L^k \otimes A^{-u})| \leq C_2 r(k+1)^{n-1}.$$

Indeed, the difference is bounded by the sum of the adjacent dimensions $h^{j-1}(Y, \mathcal{Q} \otimes L^k)$ and $h^j(Y, \mathcal{Q} \otimes L^k)$.

To remove the fixed negative twist, choose a nonzero section of A^u and let D be its zero scheme. Since Y is integral, this is an effective Cartier divisor and

$$0 \rightarrow A^{-u} \rightarrow \mathcal{O}_Y \rightarrow \mathcal{O}_D \rightarrow 0$$

is exact. No smoothness of D is needed. Formula (3.1) makes $\mathcal{O}_D$ S -regular. At twist $T = S$, the vanishing $H^1(Y, A^{T-u}) = 0$ gives

$$h^0(Y, \mathcal{O}_D \otimes A^T) \leq h^0(Y, A^T) \leq C_3,$$

where the last bound is uniform on the model. As $\dim D \leq n - 1$, [Theorem 2.2](#) and the divisor sequence imply

$$(3.5) \quad |h^j(Y, L^k \otimes A^{-u}) - h^j(Y, L^k)| \leq C_4(k+1)^{n-1}.$$

Combining [\(3.4\)](#) and [\(3.5\)](#), and using

$$h^j(Y, \mathcal{E} \otimes L^k) = h^j(Y, L^{pk}),$$

proves [\(3.2\)](#). The last equality is an equality of dimensions: the action on the ground field is twisted by its Frobenius automorphism, which preserves vector-space dimension. Frobenius has been taken after choosing the geometric fiber; no commutation of absolute Frobenius direct image with a field extension is being assumed. $\square$

4. SPECIALIZATION AND CONVERGENCE

We first control the change between consecutive powers on the fixed complex variety.

Lemma 4.1. *Let X be a connected smooth complex projective variety of dimension $n \geq 1$, and let L be a line bundle. For any fixed $j \geq 0$, the sequence*

$$a(k) = \frac{h^j(X, L^k)}{k^n}, \quad k \geq 1,$$

is nonnegative and bounded, and there is a constant C_5 such that

$$(4.1) \quad |a(k+1) - a(k)| \leq \frac{C_5}{k} \quad (k \geq 1).$$

Proof. Fix a very ample line bundle A . Applying [Theorem 2.2](#) to $\mathcal{O}_X$, with any fixed nonnegative regularity and reference twist, gives $h^j(X, L^k) \leq C_6(k+1)^n$ and hence boundedness of $a(k)$.

Choose $v \geq 1$ so that A^v and $L \otimes A^v$ are globally generated. Nonzero sections define effective Cartier divisors D_0, D_1 , respectively, because X is integral. Their sequences give

$$\begin{aligned} 0 &\longrightarrow L^k \longrightarrow L^{k+1} \otimes A^v \longrightarrow \mathcal{O}_{D_1} \otimes A^v \otimes L^{k+1} \longrightarrow 0, \\ 0 &\longrightarrow L^{k+1} \longrightarrow L^{k+1} \otimes A^v \longrightarrow \mathcal{O}_{D_0} \otimes A^v \otimes L^{k+1} \longrightarrow 0. \end{aligned}$$

Each fixed sheaf $\mathcal{O}_{D_i} \otimes A^v$ has support dimension at most $n - 1$. The proposition bounds its cohomology after tensoring with L^{k+1} by $C_7(k+1)^{n-1}$. Comparing both end terms with the common middle term gives

$$|h^j(X, L^{k+1}) - h^j(X, L^k)| \leq C_8(k+1)^{n-1}.$$

After dividing by $(k+1)^n$, the remaining change in normalization is bounded by

$$h^j(X, L^k) \left(\frac{1}{k^n} - \frac{1}{(k+1)^n} \right) \leq \frac{C_9}{k}.$$

These two bounds prove [\(4.1\)](#). $\square$

Proof of Theorem A. If $n = 0$, the variety is a finite collection of reduced points: $h^0(X, L^k)$ is constant and the higher cohomology groups vanish. For $n \geq 1$, cohomology in degrees $j > n$ vanishes. It remains to treat $0 \leq j \leq n$. We first assume that X is connected and hence integral.

Choose a model as in [Theorem 3.1](#). Put $K = \text{Frac } R$ and let η be the generic point of $\text{Spec } R$. Extension of the ground field preserves cohomology dimensions [[Stacks](#), Tag [0BDN](#)], so for every $k \geq 1$,

$$h^j(X, L^k) = \dim_K H^j(\mathcal{X}_\eta, \mathcal{L}_\eta^k).$$

Use $a(k)$ as in [Theorem 4.1](#). For a geometric fiber Y write $a_Y(k) = h^j(Y, L_Y^k)/k^n$. Dividing [\(3.2\)](#) by $(pk)^n$, and increasing one constant to absorb $(k+1)^{n-1}/k^{n-1}$, gives

$$(4.2) \quad |a_Y(pk) - a_Y(k)| \leq \frac{C_{10}}{k} \quad (k \geq 1).$$

The constant C_{10} is independent of the fiber, the prime, and k .

Fix a positive integer k . Upper semicontinuity gives a nonempty open subset $U_k \subset \text{Spec } R$ where the cohomology dimension of $\mathcal{L}^k$ equals its generic value [[Stacks](#), Tag [0BDN](#)]. Its image

in $\text{Spec } \mathbb{Z}$ is constructible by Chevalley's theorem [Stacks, Tag 054K] and contains the generic point. A constructible subset containing the generic point of this integral noetherian scheme contains a nonempty open subset. Thus U_k has a nonempty fiber over every sufficiently large prime p . Choose a closed point in that fiber and extend its finite residue field to an algebraic closure, obtaining a geometric fiber $Y_{k,p}$ with

$$a_{Y_{k,p}}(k) = a(k).$$

The prime threshold and the chosen point may depend on k .

At exponent pk , upper semicontinuity over the integral base gives the inequality $a(pk) \leq a_{Y_{k,p}}(pk)$, since the generic cohomology dimension is the minimum. Combining this with (4.2) proves

$$(4.3) \quad a(pk) \leq a(k) + \frac{C_{10}}{k} \quad \text{for every sufficiently large prime } p.$$

This step uses equality only at the fixed exponent k . Equality at the varying exponent pk is unnecessary, and no open subset working simultaneously for all exponents has been chosen.

Keep k fixed, let m tend to infinity, and take p_m to be the largest prime at most m/k . The prime number theorem implies $kp_m/m \rightarrow 1$. Indeed, for every $0 < \varepsilon < 1$, the asymptotic formula for the number of primes gives a prime in $((1 - \varepsilon)x, x]$ for all sufficiently large x . The same relative-interval consequence is used in [Tot13, proof of Theorem 6.3, p. 745]. In particular, p_m eventually exceeds the fixed- k threshold in (4.3). Summing (4.1) gives

$$|a(m) - a(kp_m)| \leq C_5 \sum_{\ell=kp_m}^{m-1} \frac{1}{\ell} \rightarrow 0,$$

because $kp_m/m \rightarrow 1$ and $kp_m \rightarrow \infty$.

The boundedness in Theorem 4.1 makes $\alpha = \limsup_{m \rightarrow \infty} a(m)$ and $\beta = \liminf_{m \rightarrow \infty} a(m)$ finite. The preceding comparison and (4.3) show that

$$(4.4) \quad \alpha \leq a(k) + \frac{C_{10}}{k} \quad (k \geq 1).$$

Choose integers $k_\nu \rightarrow \infty$ with $a(k_\nu) \rightarrow \beta$. Passing to the limit in (4.4) gives $\alpha \leq \beta$. Since $\beta \leq \alpha$ by definition, $a(k)$ converges to their common finite value.

Finally, a smooth projective variety has finitely many connected components. Each component of the pure n -dimensional X satisfies the result just proved, and cohomology dimensions add across these components. Their finite sum therefore has a limit. $\square$

Remark 4.2. For $n \geq 1$, if λ_j denotes the limit in Theorem A, (4.4) gives $h^j(X, L^k)/k^n \geq \lambda_j - C/k$ for a fixed constant C . This is a one-sided estimate; the proof does not supply an upper error bound of order $1/k$ in characteristic zero.

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